

BIOPHLX | Full app test results

September 19, 2026 - local working app plus live AWS integrations

Verdict: not ready for production sign-off

NOT READY: standalone history persistence fails; production payment safety and end-to-end validation outstanding.

Test layer	Result	What was exercised
Unit tests	12 passed	Scores, band serialization, history grouping, theme
Browser customer flow	31 checks passed	22 walkthrough steps + signup/saved workout + history checks
Live AWS	38 passed / 2 failed	Cognito, AppSync, DynamoDB, S3; failures concern one standalone-save defect

What passed

Login and token refresh; profile create/read/update; trainer and workout queries; workout/session/set/rep writes; saved-workout exercise readback; independent band snapshots; profile photo upload and public display.

The browser completed login, trainer discovery, simulated purchase, tutorial playback, band placement, countdown, sample reps, save, calendar/history review, creating a workout, and dark mode. Additional checks covered signup/profile completion and performing a saved workout.

Test boundaries

- Browser uses simulated authentication, AWS, BLE and checkout; live AWS tested separately.
- No physical device, native iOS/Android build, signup email or reset-email delivery test.
- Stripe live configuration inspected only; no payment or webhook-to-entitlement transaction.

Temporary test accounts, profile image and records were cleaned up. No payment charged, production code pushed, or backend configuration changed during this test.

Failures and recommended fixes

1. Standalone exercises lose their session exercise record

Reproduced live: createSessionItem with no workout_id returns null; listSessionItemsBySession then returns no exercise. The same operation with a workout_id succeeds. Reps save independently, leaving incomplete history.

Likely cause confirmed by configuration inspection: the resolver writes workout_id as DynamoDB NULL, while workout_id is a secondary-index key. Its response mapping returns only ctx.result and does not surface ctx.error. The app also does not validate the returned exercise record.

Fix: omit optional indexed attributes when absent; propagate resolver errors; require the app to confirm a session item was saved before reporting success. Retest standalone and saved workouts.

2. Checkout trusts client-supplied price and buyer ID

Deployed source inspection found checkout price and buyer ID are taken from request arguments. The handler computes Stripe unit_amount from that price. This was not exploited, and no paid transaction was attempted.

Fix: fetch price/product ownership server-side and derive the buyer from the authenticated identity. Then test Stripe sandbox checkout, webhook verification, entitlement creation and duplicate-event handling.

Still unverified

Native mobile execution and real band behavior; actual signup/reset email delivery; paid checkout and webhook-to-workout access. Passing the browser simulation does not verify these integrations.

Live AWS result details

PASS - App-config API key: exercise catalog

Live response matched expected contract.

PASS - Unauthenticated request rejected

Request denied without authentication

PASS - Cognito password sign-in

Temporary test account authenticated; messages suppressed.

PASS - Cognito token refresh

Live response matched expected contract.

PASS - Cognito wrong-password rejection

Incorrect password rejected.

PASS - Cognito create User

Live response matched expected contract.

PASS - Cognito create Customer

Live response matched expected contract.

PASS - Cognito create Workout

Live response matched expected contract.

PASS - Profile read after create

Live response matched expected contract.

PASS - Profile update

Live response matched expected contract.

PASS - Cognito Dashboard sessions

Live response matched expected contract.

PASS - API-key authorization probe Dashboard sessions

Expected denial: protected data requires a Cognito user token.

PASS - Cognito Saved workout library

Live response matched expected contract.

PASS - API-key authorization probe Saved workout library

Expected denial: protected data requires a Cognito user token.

PASS - Cognito Trainer directory

Live response matched expected contract.

PASS - API-key authorization probe Trainer directory

Expected denial: protected data requires a Cognito user token.

PASS - Cognito Purchased permissions

Live response matched expected contract.

PASS - API-key authorization probe Purchased permissions

Expected denial: protected data requires a Cognito user token.

PASS - Cognito exercise catalog

Live response matched expected contract.

PASS - Create WorkoutItem

Live response matched expected contract.

PASS - Create Session

Live response matched expected contract.

FAIL - Create SessionItem

Stronger follow-up assertion found null mutation result and no saved standalone exercise record.

PASS - Create SessionItemSet

Live response matched expected contract.

PASS - Create SessionItemRep

Live response matched expected contract.

PASS - Read saved workout items

Live response matched expected contract.

PASS - Read rep metrics and combined score

Live response matched expected contract.

PASS - Read recorded session on dashboard

Live response matched expected contract.

PASS - Trainer products query contract

Live response matched expected contract.

PASS - Cognito sign-out

Live response matched expected contract.

PASS - Cognito identity exchange

Authenticated identity issued temporary storage credentials.

PASS - S3 profile upload with app identity

Unique temporary account profile path accepted.

PASS - S3 app public image URL

Uploaded bytes returned from the app URL pattern.

PASS - Stripe Lambda configuration

State=Active; key modes={"STRIPE_SECRET_KEY": "live"}

PASS - Create schema-compatible rep 1

Live response matched expected contract.

PASS - Create schema-compatible rep 2

Live response matched expected contract.

PASS - Both bands persist independently

Live response matched expected contract.

PASS - Read saved scores and Float metrics

Live response matched expected contract.

PASS - Saved-workout session item create

Live response matched expected contract.

PASS - Saved-workout session item readback

Live response matched expected contract.

FAIL - Standalone session exercise readback

No exercise returned after createSessionItem without a workout ID.

Browser and unit evidence

All browser data, band signals and checkout events were simulated. No browser runtime errors occurred in the completed flows.

PASS - Welcome to BIOPHLX

PASS - Sign in and get started

PASS - Meet your dashboard

PASS - Find your trainer

PASS - Explore a workout

PASS - Purchase with confidence

PASS - Your workout library

PASS - Learn how to wear the bands

PASS - Connect both bands

PASS - Choose an exercise

PASS - Confirm your placement

PASS - Hold still, then begin

PASS - See every rep

PASS - Explore your rep history

PASS - Review and save

PASS - See where the work went

PASS - Choose a day on the calendar

PASS - Open that day's performance

PASS - Compare your training

PASS - Create your own plan

PASS - Ready for your next session

PASS - Your progress, in your style

PASS - Signup, confirmation, login and profile completion (mock auth)

PASS - Perform saved workout, persist rep and both band snapshots (mock AWS/BLE)

PASS - No browser runtime errors in extended flows

PASS - Calendar date filters muscle chips and chart

PASS - Set expands to individual rep scores

PASS - Rep expands to separate left and right metrics

PASS - Light and dark rendering

PASS - Legacy records show unavailable band data

PASS - Empty date does not borrow another day

Unit suite: 12 passed, 0 failed. Raw results: unit-results.tap. Structured results: test-results.json. Original backend observations and screenshots are retained beside this PDF.

Harness corrections: fixed a copied Unicode selector and disambiguated repeated exercise/Perform labels. API-key denials are counted as expected authorization passes. A weak create-only check was superseded by a null-result/readback failure assertion.